Lesson 3 — Regression

Technologies for Artificial Intelligence

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Before we start

- Exercise 2 was due today
- The pipeline you built is the one we fit models into from now on

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Today: the first real model

- The model and what it claims
- The exact solution — and why it sometimes fails
- Gradient descent
- Curves, and the price of flexibility
- Ridge and Lasso

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One dataset, all lesson

600 houses. Six measurements. A price.

And we know the coefficients that generated it.

What a linear model claims

Each feature contributes a fixed amount per unit, and the contributions add up.

- Every square metre: the same 2,400 €
- Every kilometre out: the same −6,500 €

Fitting means minimising something

We need one number saying how badly a candidate model is doing.

Adding up the errors fails: +50,000 and −50,000 cancel to zero.

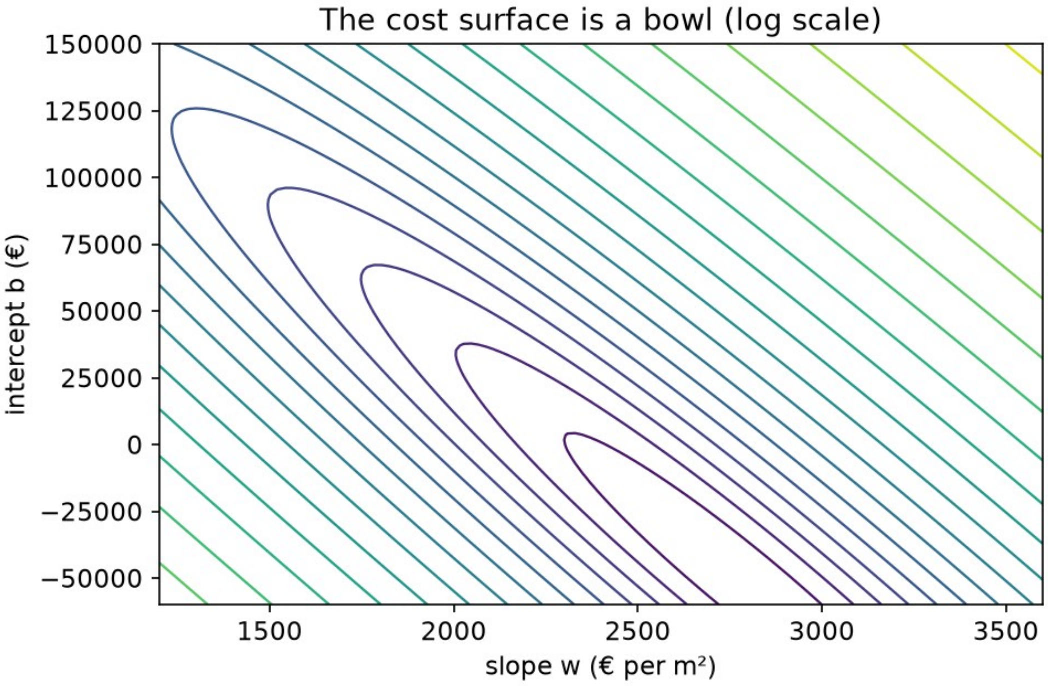
Why squared, not absolute

- **Differentiable everywhere** — no corner at zero
- **Punishes one large error more** than several small ones
- Under Gaussian noise, it **is** maximum likelihood

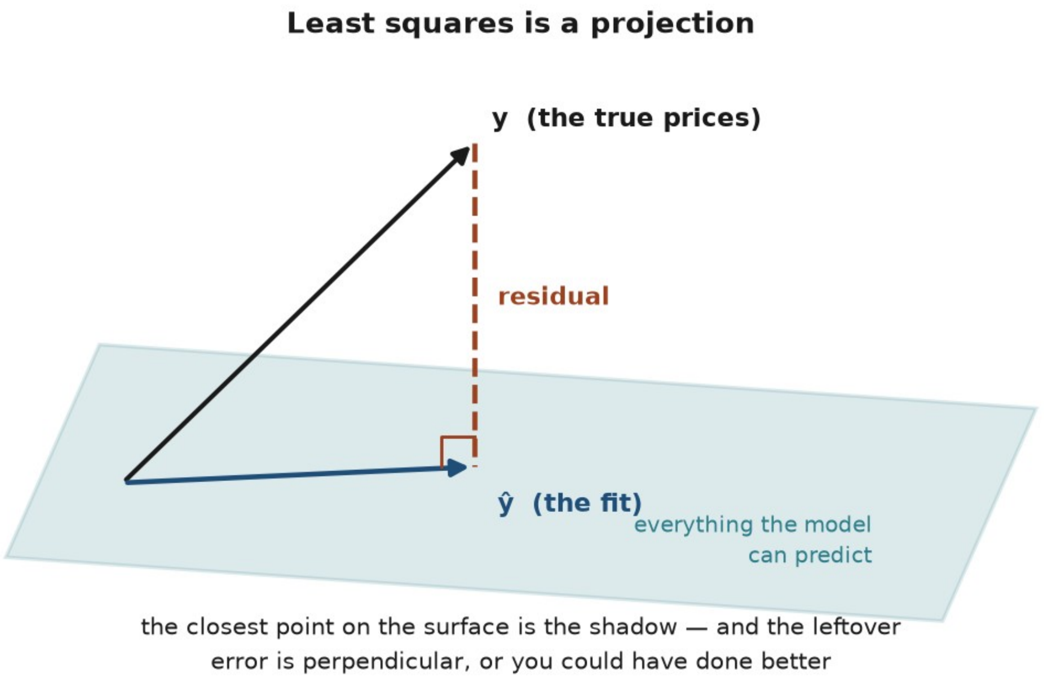
The cost function — mean squared error

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^{m} \left(\hat{y}^{(i)} - y^{(i)} \right)^2$$

The cost is a bowl



The picture behind the exact solution



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The normal equation

$$\theta = (X^\top X)^{-1} X^\top y$$

So why learn anything else?

- $(X^T X)^{-1}$ costs about n^3 operations
- Two columns carrying one fact → **no inverse exists**
- Nearly-redundant columns → an inverse that is useless

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The picture behind gradient descent

You are on a hillside in fog.

You cannot see the valley — but you can feel which way the ground slopes.

The update rule

$$w \leftarrow w - \alpha \frac{\partial J}{\partial w}, \qquad \frac{\partial J}{\partial w_j} = \frac{1}{m}\sum_i \left(\hat{y}^{(i)} - y^{(i)}\right) x_j^{(i)}$$

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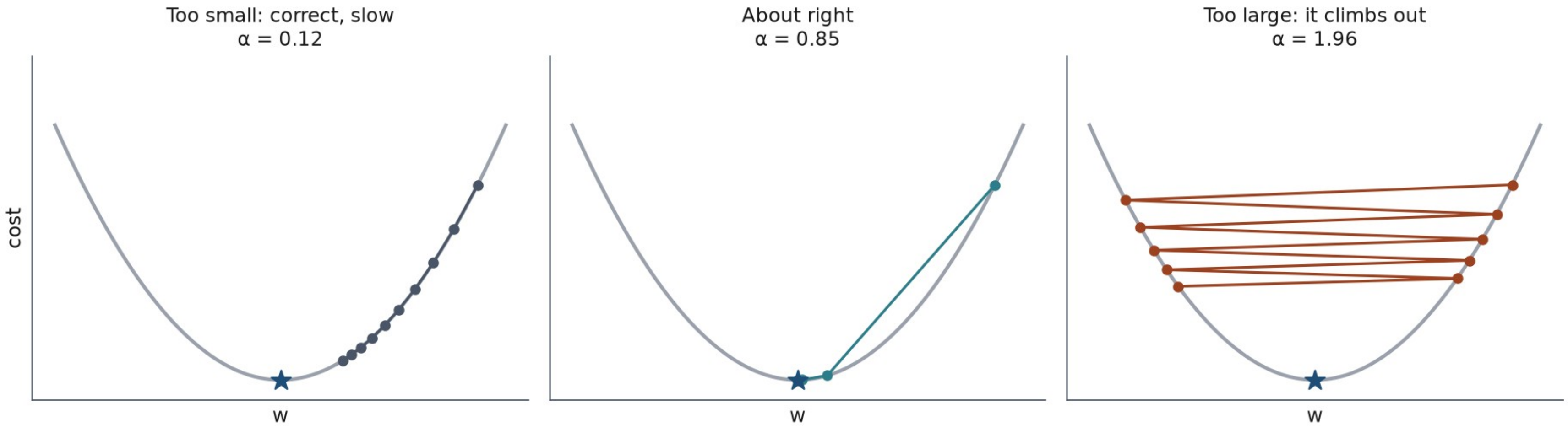
Three learning rates



Notebook 1, live

The model, the cost, the exact solution, the iterative one — from scratch.

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Break

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What if the relationship bends?

Energy consumption against temperature: heating in the cold, cooling in the heat.

No straight line follows that.

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"Linear" means linear in the coefficients

$$\hat{y} = w_1 t + w_2 t^2 + b$$

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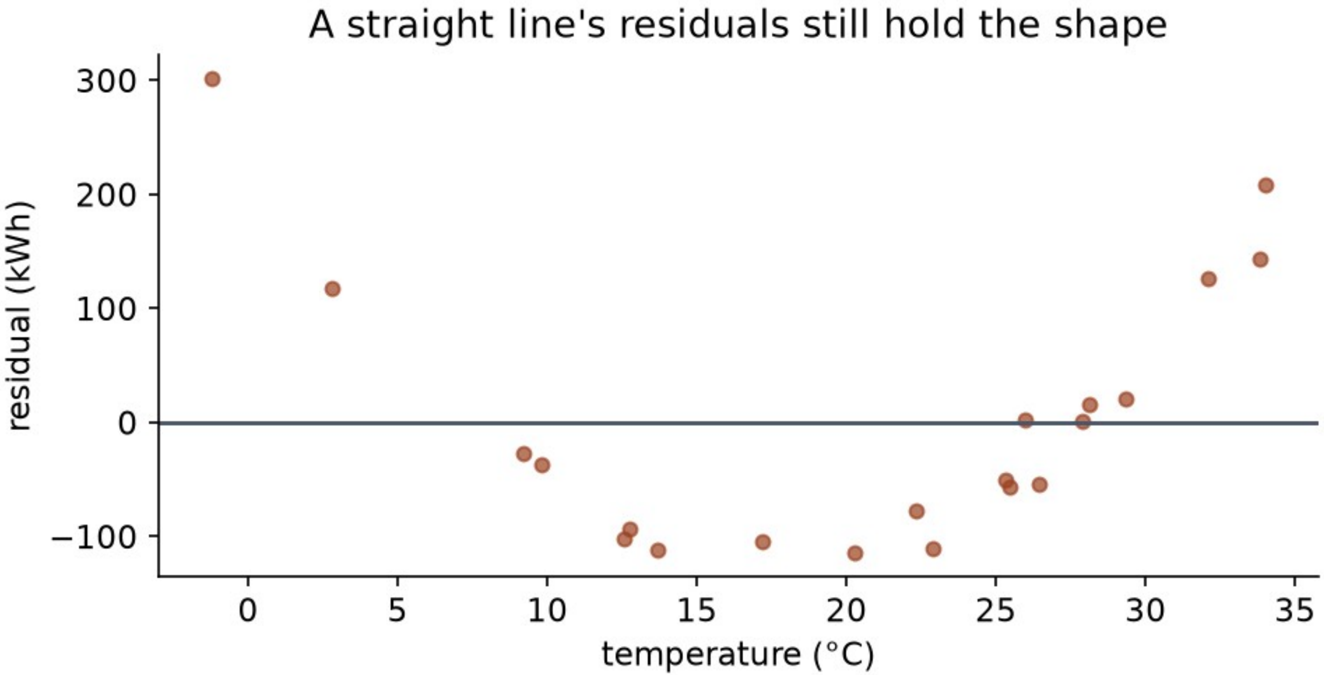
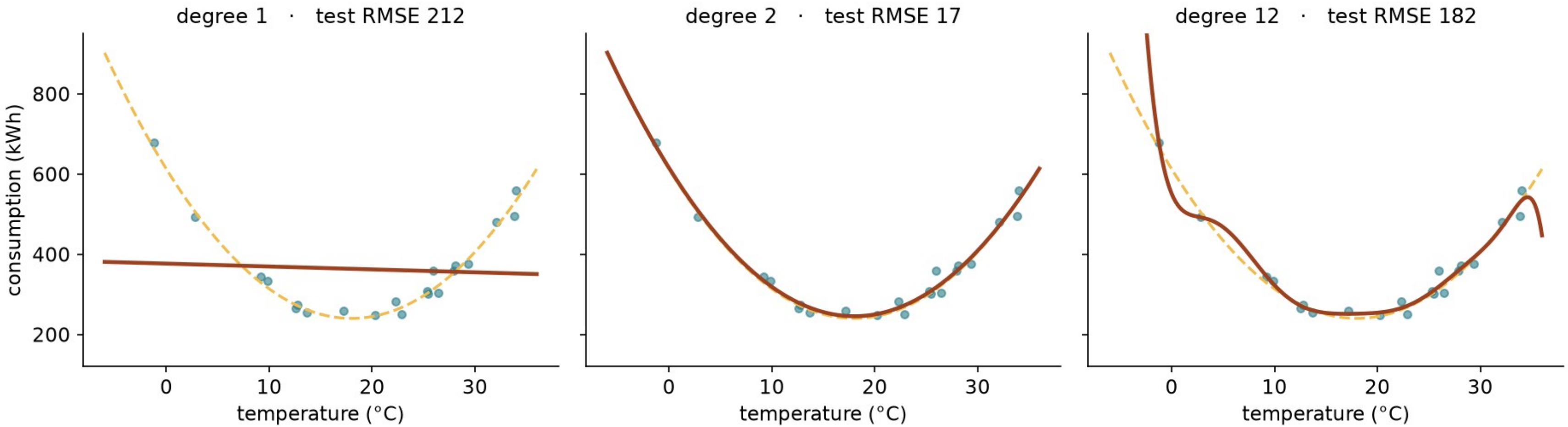
The straight line’s residuals keep the shape



So use more flexibility?



The table tha

Degree	Train RMSE	Test RMSE
1	113.3	212.1
3	17.5	**16.9**
9	16.5	118.2
12	16.3	182.0

Root mean squared
of the MSE, so it reads in the units of the target:

Why flexibility goes wrong

Largest coefficient, degree 2: **411**

Largest coefficient, degree 12: **247,514**

Charge for size

$$J_{\text{ridge}} = \text{MSE} + \lambda \sum_j w_j^2 \qquad J_{\text{lasso}} = \text{MSE} + \lambda \sum_j |w_j|$$

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What a penalty buys

Model	Train	Test	max |w|
none	16.3	182.0	247,514
λ = 0.01	18.0	**22.8**	365
λ = 1	44.5	105.2	156
λ = 100	96.0	227.6	6

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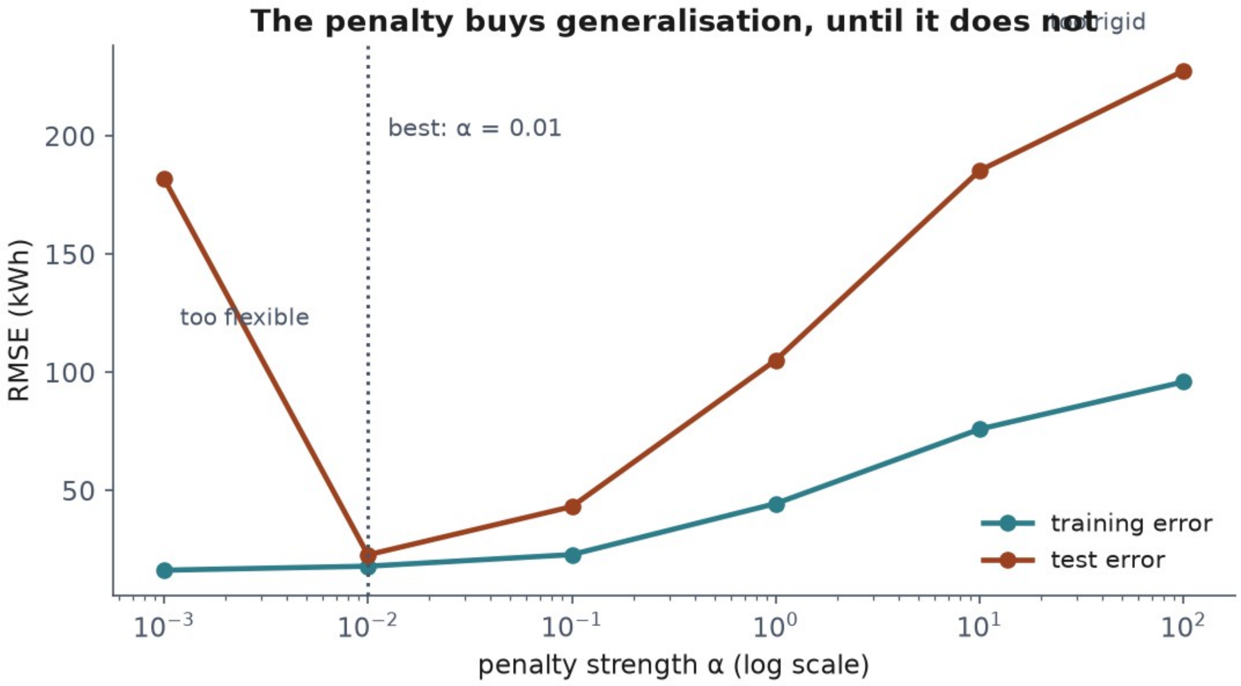
Too flexible, too rigid



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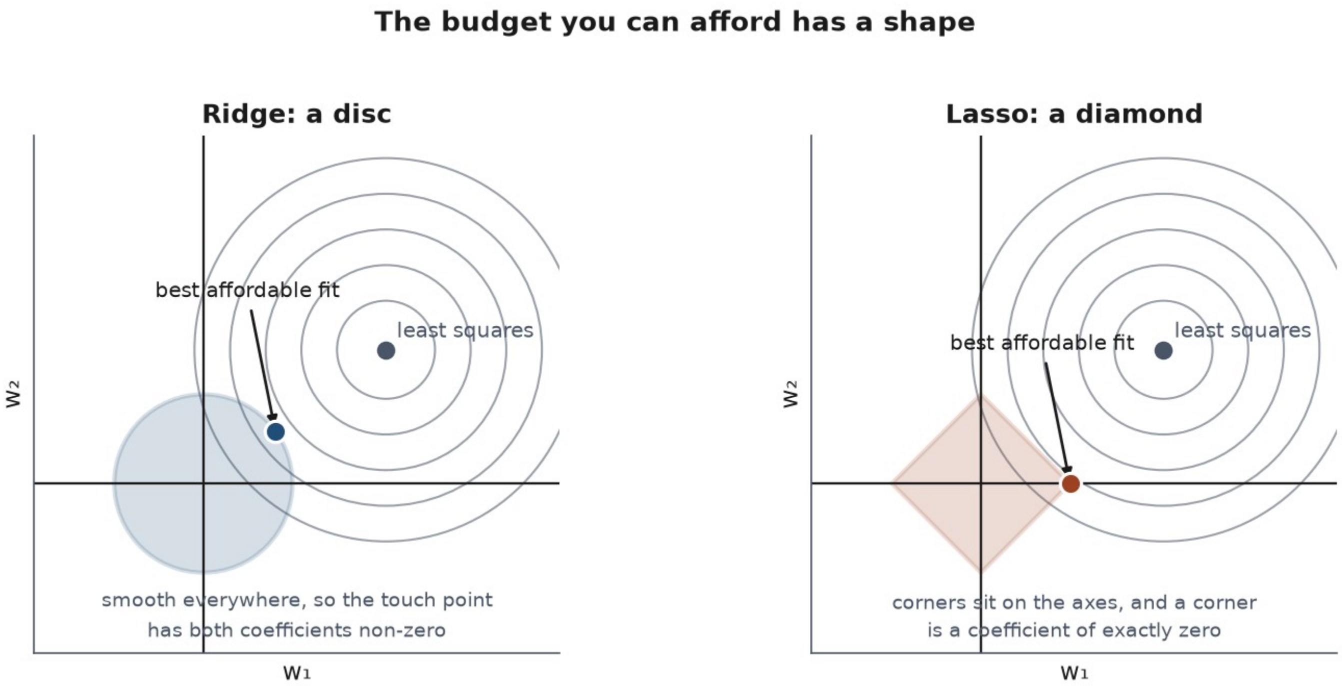
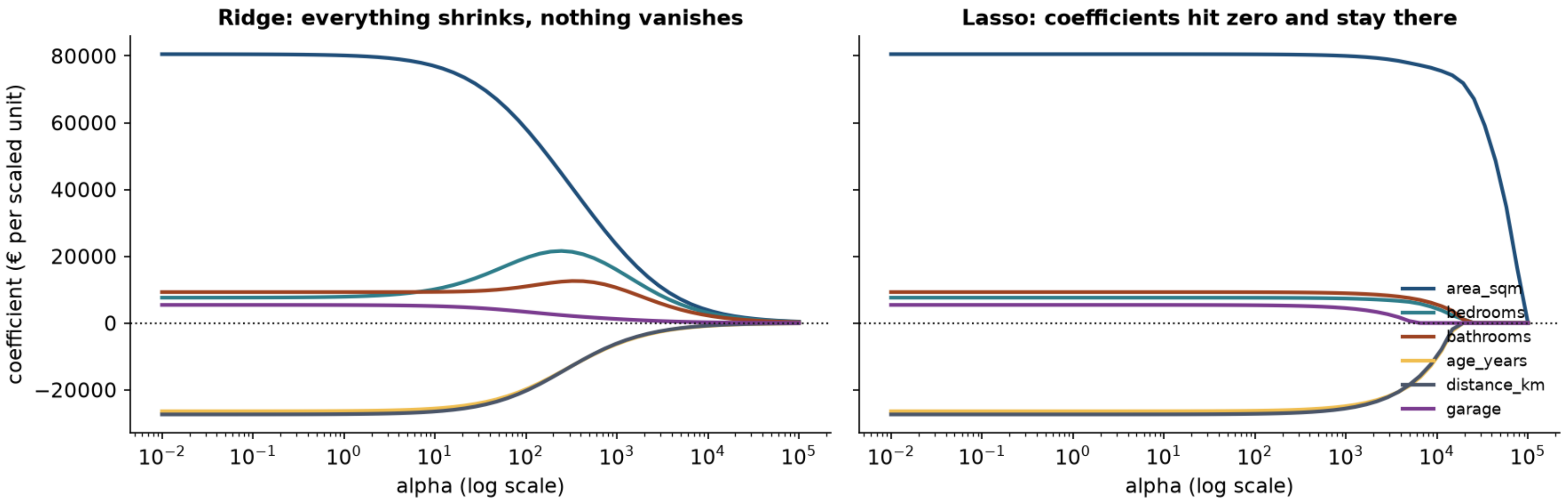
The picture behind Ridge vs Lasso



Ridge shrinks, Lasso selects



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The case Ridge was invented for

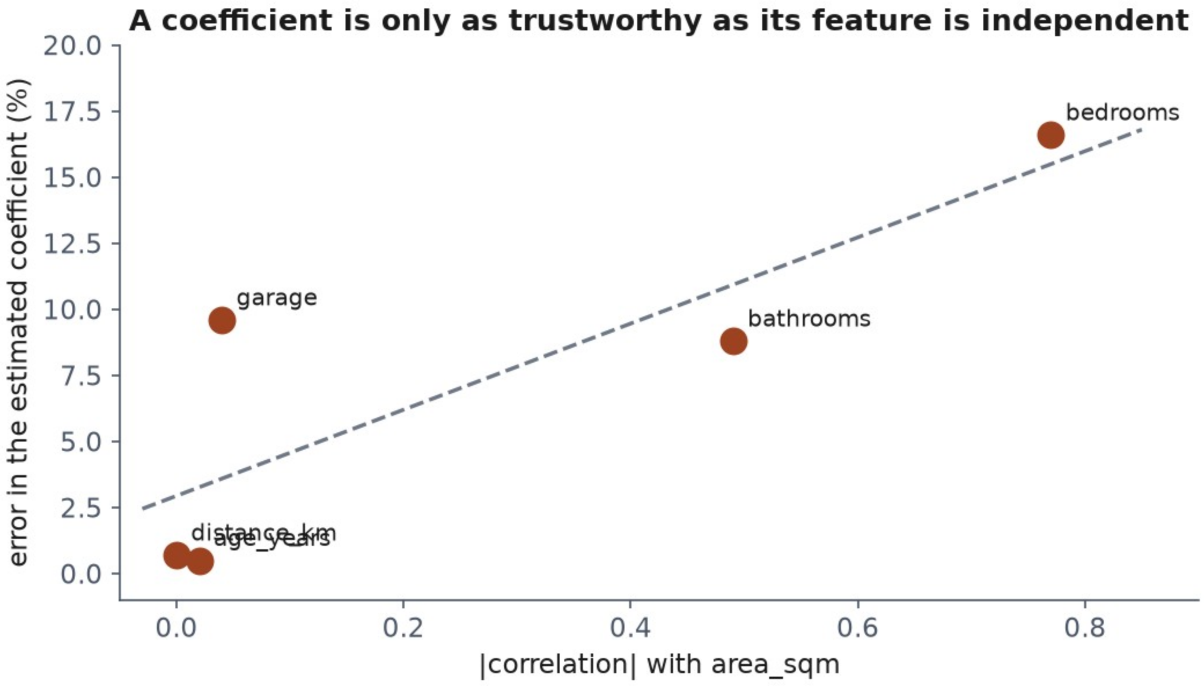
`area_sqm` and `area_sqft` — the same measurement, two units.

Correlation: **0.999999**

The coefficients are noise

Split	area_sqm	area_sqft
0	9,261	−640
1	14,535	−1,127
2	45,307	−3,989
3	39,061	−3,402

When can you trust a coefficient?



Notebook 3, live

Ridge on the disaster, the regularisation paths, the collinear case.

What we did today

- A model that claims a fixed amount per unit
- An exact solution — a projection — and when it fails
- An iterative one: walk downhill, mind your stride
- Flexibility helps until it does not
- Charging for size buys generalisation, and readable coefficients

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Homework — due Friday 16 October

`Exercises/03_regression.md`

Fit, regularise, and **explain which coefficients you believe**.

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